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Isolation and identification of neuroprotective compounds from *Morus alba* L. root bark against NMDA-induced retinal degeneration

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ABSTRACT

Morus alba L. root bark (MAR) has been used in traditional Chinese medicine and has recently attracted attention for its neuroprotective potential. In this study, 23 compounds were isolated from MAR, including two new compounds (**1** and **2**) and six new natural products (**3–8**) present as stereoisomeric mixtures, all characterised by NMR, ECD, and HRESIMS. Bioactivity screening identified five compounds (**2**, **9**, **11**, **18**, and **19**) with significant neuroprotective effects, with compound **19** ($EC_{50} = 0.3 \pm 0.12 \mu M$) showing the strongest effect. Under glutamate-induced oxidative stress, compound **19** reduced ROS levels from 2.23-fold to near control levels (1.07-fold) and markedly decreased apoptosis. In an NMDA-induced retinal degeneration mouse model, compound **19** preserved retinal thickness, reduced retinal ganglion cell loss, and suppressed apoptosis-related protein expression. These results significantly expand the phytochemical profile of MAR and highlight compound **19** as a promising glaucoma therapeutic candidate.

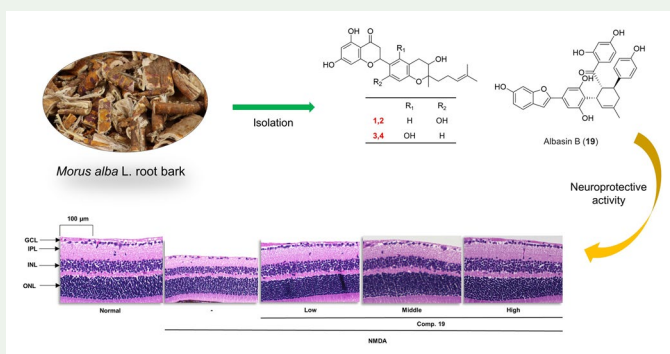
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1. Introduction

Glaucoma is one of the leading causes of irreversible blindness worldwide and is characterised by the progressive degeneration of retinal ganglion cells (RGCs) (Lee and Mackey 2022; Qi et al. 2025). Current therapeutic strategies mainly focus on reducing intraocular pressure (IOP), the major modifiable risk factor for the disease (Liu et al. 2022; Sharif 2023). However, despite effective IOP control, the progressive loss of RGCs and visual field deterioration often persist in many patients (Le et al. 2021; Wang et al. 2024). This observation suggests that additional pathogenic mechanisms beyond elevated IOP contribute to glaucomatous neurodegeneration. Among these mechanisms, glutamate-induced excitotoxicity has been recognised as a key contributor to RGC death (Neves et al. 2023; Wu et al. 2025). Excessive extracellular glutamate triggers overactivation of glutamate receptors, resulting in intracellular calcium overload, oxidative stress, and ultimately neuronal apoptosis (Kang et al. 2021; Wu et al. 2025). To date, no approved pharmacological agent effectively protects RGCs from degeneration in glaucoma (Wang et al. 2024). Therefore, developing neuroprotective strategies targeting oxidative stress and excitotoxicity is of great importance for preventing glaucomatous vision loss.

Morus alba L. is well known for its wide range of pharmacological activities (Li, Cao et al. 2025; Morales Ramos et al. 2021). Every organ of this plant has been investigated for its medicinal potential and use in functional food production (Batiha et al. 2023). In particular, the root bark of *M. alba* (MAR) has been traditionally used in Chinese medicine for the treatment of diabetes, inflammation, hypertension, and hypoglycaemia. The root bark is typically administered in the form of water decoctions, powdered preparations, or herbal formulas such as Sang-Bai-Pi Tang, where it is combined with other medicinal plants to alleviate pulmonary heat, reduce swelling, and promote diuresis. Its use is documented in classical materia medica texts and remains common in modern clinical practice (Culenova et al. 2020; Li et al. 2025). Recent studies have reported its neuroprotective effects in Alzheimer's disease models, in which cognitive decline is associated with progressive neurodegeneration (Xia et al. 2019). Extensive phytochemical studies on MAR have reported a wide variety of secondary metabolites, including Diels–Alder-type adducts, prenylated flavonoids, benzofurans, chalcones, flavanones, and various polyphenolic constituents (Du et al. 2003; Culenova et al. 2020; Tan et al. 2025; Le et al. 2026). These compounds have demonstrated diverse biological activities such as anti-inflammatory, antioxidant, antiviral, and cardioprotective activities in several *in vitro* models (Du et al. 2003; Zheng et al. 2017; Tan et al. 2025). Despite these extensive phytochemical and pharmacological investigations, the neuroprotective effects of MAR constituents on RGCs and their potential relevance to glaucoma-associated excitotoxicity remain poorly understood and have not been systematically explored. Therefore, this study aimed to isolate and characterise the bioactive constituents from MAR and evaluate their neuroprotective effects on RGCs using both *in vitro* oxidative stress assays and an *in vivo* NMDA-induced retinal degeneration model.

2. Results and discussion

2.1. Structural elucidation of compounds 1–23

The trivial names morutamins M–T (compounds **1–8**) were assigned based on the plant source (*Morus alba*) in accordance with established natural products

nomenclature and as a continuation of the previously reported morutamin series (A–L) from *M. alba* root bark (Le et al. 2026).

Morutamins M (**1**) and N (**2**) were both yielded as yellowish powders, both exhibiting the molecular formula of $C_{25}H_{28}O_7$ as determined from their HRESIMS data (m/z 463.1730 for **1** and 463.1731 for **2** [$M+Na$] $^+$, calcd. for $C_{25}H_{28}O_7Na^+$, 463.1733). The 1H and ^{13}C -NMR data of **1** and **2** (Tables S1 and S2) showed remarkable similarity, suggesting that they are likely stereoisomers. Because **1** and **2** share the same absolute configuration at C-2 but differ at other stereogenic centres, as discussed below, they are classified as diastereomers. 1H NMR spectra of **1** and **2** displayed three signals at [δ_H 2.72 and 3.19 (each 1H, dd) and 5.70 (1H, dd)] for **1** and [δ_H 2.72 and 3.18 (each 1H, dd) and 5.70 (1H, dd)] for **2**, corresponding to H₂-3 and H-2 of the flavanone skeleton (Asai et al. 2008); two aromatic proton signals at δ_H 6.37 (1H, s, H-3') and 7.19 (1H, s, H-6') for both compounds indicated *para*-substituted protons on ring B. These characteristics established a flavanone backbone bearing *para*-substituted aromatic protons on ring B of **1** and **2**. Analysis of the HMBC and 1H - 1H COSY spectra (Figure S65A) demonstrated the presence of a pyran ring involving C-4', C-5', C-9, C-10, and C-11, linked by an oxygen bridge between C-4' and C-11, as supported by HMBC correlations from H-9 to C-4'/C-6'/C-11 and 1H - 1H COSY correlation between H-9 and H-10. The downfield chemical shifts at δ_C 68.64 for **1** and 68.60 for **2** confirmed oxygen substitution at C-10. In addition, HBMBC correlations from H-12 to C-10/C-13 suggested the positions of the methyl group and 2-methylhex-2-ene at C-11 (Figure S65A). NOESY correlations of both compounds between H-9a/H-10/H-13 and H-9b/H-12 (Figure S65B) supported the assignment of the relative stereochemistry at C-10 and C-11 as *rel*-10*S*,11*R*. Analysis of the ECD spectra of **1** and **2** (Figure S65C) revealed positive Cotton effects at approximately 330 nm and negative ones near 295 nm, allowing the absolute configuration at C-2 to be assigned as *S* (Zheng et al. 2012). Taken together, their structures were shown in Figure 1.

Morutamins O–T (**3–8**) were obtained as chromatographically homogeneous fractions, each appearing as a single peak on reversed-phase HPLC analysis in multiple purification trials. However, the near-baseline ECD spectra of **3** and **4**, together with the duplicated sets of 1H and ^{13}C NMR resonances observed for **5–8** (see detailed discussion below), indicate that these samples consist of inseparable stereoisomeric

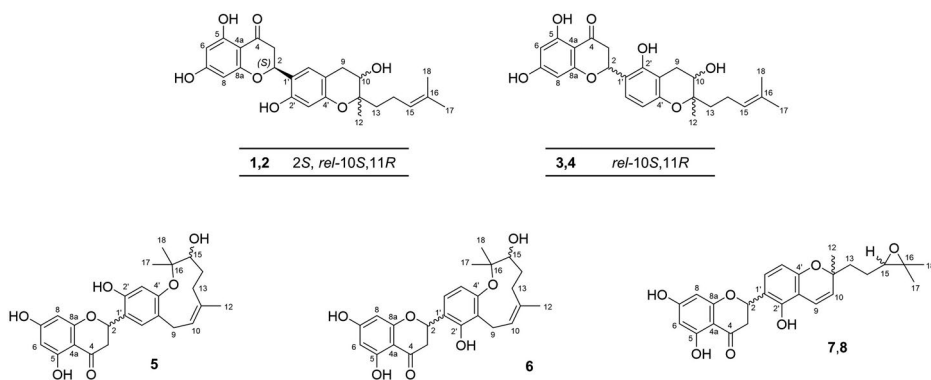


Figure 1. Structures of new isolates **1–8** from *M. alba* root bark.

mixtures with identical molecular formulas rather than single stereochemically defined compounds. Due to the limited amounts available and partial degradation observed during storage, further chromatographic separation of the individual stereoisomers could not be achieved. Accordingly, compounds **3–8** are described here as new natural product stereoisomeric mixtures. Because these samples were not obtained as single stereochemically defined compounds, they were not subjected to biological evaluation in this study.

Morutamins **O (3)** and **P (4)**, similar to **1** and **2**, were characterised as a pair of stereoisomers. Their ^1H and ^{13}C -NMR data (Tables S1 and S2), along with HRESIMS data (m/z 463.1735 for **3** and 463.1738 for **4** [$\text{M}+\text{Na}$] $^+$, calcd. for $\text{C}_{25}\text{H}_{28}\text{O}_7\text{Na}^+$, 463.1733), confirmed that both compounds share an identical 2D structural framework and the molecular formula $\text{C}_{25}\text{H}_{28}\text{O}_7$ and closely resembled those of **1** and **2**. The difference lies on the presence of *ortho*-positioned protons of **3** and **4** instead of *para*-positioned protons of **1** and **2** on ring B at [δ_{H} 6.43 (1H, d, $J=8.6\text{ Hz}$, H-5') and 7.22 (1H, d, $J=8.6\text{ Hz}$, H-6')] for **3** and [δ_{H} 6.44 (1H, d, $J=8.6\text{ Hz}$, H-5') and 7.23 (1H, d, $J=8.6\text{ Hz}$, H-6')] for **4**. Similarly, NOESY correlations of both compounds between H-9a/H-10/H-13 and H-9b/H-12 (Figure S65B) supported the assignment of the relative stereochemistry at C-10 and C-11 as *rel*-10*S*,11*R*. However, the ECD spectra of **3** and **4** displayed no discernible Cotton effects and fluctuated around the baseline, suggesting that these compounds may exist as racemates or enantiomeric mixtures. Consequently, the absolute configurations at C-2 of these compounds could not be assigned based on the present data. Collectively, structures of **3** and **4** were shown in Figure 1.

Morutamin **Q (5)** was yielded as a yellowish gum, and its molecular formula of $\text{C}_{25}\text{H}_{28}\text{O}_7$ was established from the protonated molecular ion peak at m/z 441.1914 [$\text{M}+\text{H}$] $^+$ (calcd. for $\text{C}_{25}\text{H}_{29}\text{O}_7^+$, 441.1913) in the HRESIMS positive ion mode. Similar to **1** and **2**, ^1H NMR spectrum of **5** revealed the presence of a flavanone backbone bearing *para*-substituted aromatic protons on ring B (Table S3). In addition, the ^1H -NMR spectrum of **5** showed the signals of one vinyl proton signal at δ_{H} 5.38 (1H, t, $J=7.3\text{ Hz}$, H-10); three methyl signals at δ_{H} 1.71 (3H, s, H-12) and 1.10 (6H, s, H-17, 18); three methylene signals at δ_{H} 3.27 (2H, overlapped, H-9), 2.32/2.10–2.04 (each 1H, m, H-13), and 1.43–1.34/1.76–1.65 (each 1H, m, H-14); and one methine signal at δ_{H} 3.28 (1H, overlapped, H-15). The ^{13}C -NMR data of **5** revealed 25 carbon signals, of which 15 were assigned to the flavanone moiety, while the remaining 10 were attributed to the monoterpene moiety. In the HMBC spectrum, the long-range correlations from H-9 to C-6'/4' located the position of the monoterpene moiety at C-5' (Figure S66). The ^1H – ^1H COSY data and HMBC correlations (Figure S66) indicated that C-9, C-10, C-11, C-13, C-14, C-15, C-16, and C-4' belong to a cyclodecane moiety, as supported by HMBC correlations from H-9 to C-4'/C-6'/C-11, from H-10 to C-5'/C-12/C-13, and from H-17 to C-15/C-16, together with ^1H – ^1H COSY correlations of H-9/H-10 and H-13/H-14/H-15. The downfield resonances at δ_{C} 79.14/79.08 confirmed oxygen substitution at C-15. As a result, the structure of **5** was characterised as shown in Figure 1.

Morutamin **R (6)** was obtained as a yellowish gum with the molecular formula of $\text{C}_{25}\text{H}_{28}\text{O}_7$, as determined by the HRESIMS ion peak at m/z 441.1912 [$\text{M}+\text{H}$] $^+$ in positive ion mode (calcd. for $\text{C}_{25}\text{H}_{29}\text{O}_7^+$, 441.1913). The ^1H -NMR and ^{13}C -NMR data (Tables S1 and S3), together with the HRESIMS data, indicated that **6** shared a close structural similarity with **5**. The key difference was the presence of *ortho*-coupled aromatic

protons in **6**, rather than the *para*-substituted protons observed in **5**, as evidenced by signals at δ_{H} 6.52 (1H, d, $J=8.4$ Hz, H-5') and 7.14 (1H, d, $J=8.4$ Hz, H-6') on ring B. Accordingly, the structure of **6** was elucidated as shown in Figure 1.

Morutamin S (**7**) was isolated as a yellowish gum. Its molecular formula was determined as $\text{C}_{25}\text{H}_{26}\text{O}_7$ by the HRESIMS ion peak at m/z 439.1759 $[\text{M}+\text{H}]^+$ in positive ion mode (calcd. for $\text{C}_{25}\text{H}_{27}\text{O}_7^+$, 439.1757). Similar to **6**, ^1H NMR spectrum of **7** revealed the existence of a flavanone backbone bearing *ortho*-substituted aromatic protons on ring B (Table S3). In addition, the ^1H -NMR spectrum of **7** showed the signals of a *gem*-dimethyl group at δ_{H} 1.11 (6H, s, H-17, 18); one methyl group δ_{H} 1.37 (3H, s, H-12); an AB spin system at δ_{H} 6.81 (1H, d, $J=10.0$ Hz, H-9) and 5.69 (1H, d, $J=10.0$ Hz, H-10); two methylene signals at δ_{H} 1.73–1.61/2.09–2.00 (each 1H, m, H-13) and 1.78–1.68 (2H, m, H-14); and one methine signal at δ_{H} 3.26 (1H, overlapped, H-15) (Table S3). HMBC correlations from H-9 to C-2'/C-4', from H-12 to C-10/C-13, and from H-15 to C-13/C-17 confirmed positions of the methyl group, pyran ring, and 2-methylhexane (Figure S66). In addition, the downfield chemical shifts at δ_{C} 79.83 and 73.41 indicated oxygen substitution at C-15 and C-16, suggesting the formation of an ether linkage (C15–O–C16). Consequently, its structure was shown in Figure 1.

Morutamin T (**8**) was obtained as a yellowish gum with the molecular formula of $\text{C}_{25}\text{H}_{26}\text{O}_7$ from its HRESIMS ion peak at m/z 439.1759 $[\text{M}+\text{H}]^+$ in positive ion mode (calcd. for $\text{C}_{25}\text{H}_{27}\text{O}_7^+$, 439.1757). The NMR data (Tables S1 and S3), together with HRESIMS analysis, indicated that **8** has the same 2D structure as **7**, with a molecular formula of $\text{C}_{25}\text{H}_{26}\text{O}_7$. As mentioned above, ^1H and ^{13}C NMR spectra of **8** (Figures S58 and S59) showed duplicated signals, indicative of stereoisomeric mixtures. The structure of **8** was shown in Figure 1.

Comparison of the physical and spectroscopic data of the isolated compounds with reported values in the literature allowed the identification of the known isolates as mulberrofuran G (**9**) (Fukai et al. 1985), 5'-(1'',1''-dimethylallyl)-5,7,2',4'-tetrahydroxyflavone (**10**) (Guo et al. 2018), soroccal B (**11**) (Qin et al. 2015), steppogenin (**12**) (Li et al. 2018), moracin P (**13**) (Lee et al. 2007), 2,2',4,4'-tetrahydroxychalcone (**14**) (Jun et al. 2007), moracenin A (**15**) (Hirakura et al. 1985; Chajra et al. 2020), moracenin B (**16**) (Chajra et al. 2020), moracenin C (**17**) (Oshima et al. 1980), sanggenon C (**18**) (Qi et al. 2016), albasin B (**19**) (Huang et al. 2017), kuwanon N (**20**) (Hano et al. 1984), kuwanon W (**21**) (Hirakura et al. 1985), sanggenon G (**22**) (Zuo et al. 2019), sanggenon X (**23**) (Lu et al. 2018) (Figure S67). These known compounds have been previously reported from members of the *Morus* genus, and their presence here aligns with the established phytochemical characteristics of mulberry species.

2.2. Evaluation of biological activities

Excess extracellular glutamate is a key contributor to RGC degeneration in glaucoma through both oxidative stress and NMDA receptor-mediated excitotoxicity. In this study, glutamate-treated R28 cells were used to model redox-dependent neuronal injury *via* glutathione depletion and reactive oxygen species (ROS) accumulation, whereas the intravitreal NMDA model was employed to mimic receptor-driven Ca^{2+} overload and apoptotic RGC death *in vivo*. These complementary models represent mechanistically linked pathways that converge on RGC loss and are widely used to

recapitulate core features of glaucomatous neurodegeneration (Tezel 2006; Dong et al. 2009).

2.2.1. Neuroprotective effects ethyl acetate fraction (MAEA) and its isolated compounds in R28 cells against glutamate-induced damage

The neuroprotective effects of the MAEA and its isolated compounds were evaluated in R28 retinal precursor cells subjected to glutamate-induced excitotoxicity. *N*-acetyl-L-cysteine (NAC, 1 mM), a well-established antioxidant, was used as a positive control. Screening across all extract fractions and isolated compounds revealed that the MAEA fraction and compounds **2**, **9**, **11**, **18**, and **19** possessed significant neuroprotective activity (Figures S68 and S69), and thus were selected for further biological evaluation. The purity of compounds **2**, **9**, **11**, **18**, and **19** used in the bioassays was confirmed to be $\geq 96\%$ by analytical HPLC analysis (Table S4).

The neuroprotective activities of the selected compounds (**2**, **9**, **11**, **18**, and **19**) were summarised in Table S5. EC_{50} values were determined only for compounds that exhibited clear, concentration-dependent neuroprotective effects (compounds **2**, **9**, **11**, **18**, and **19**, as well as the positive control NAC); the remaining compounds did not show significant protective activity within the tested concentration range, and therefore reliable EC_{50} values could not be calculated. All compounds exhibited concentration-dependent protection against glutamate-induced cytotoxicity. The concentration ranges used for EC_{50} determination were selected based on preliminary dose-screening experiments, in which compounds exhibiting higher apparent potency were evaluated at lower concentration ranges, whereas less potent compounds required higher starting concentrations to achieve measurable neuroprotective effects. Specifically, compound **2** (50–3.125 μ M), compound **9** (2–0.125 μ M), compound **11** (5–0.3125 μ M), compound **18** (10–0.625 μ M), and compound **19** (1–0.0625 μ M) were tested to determine their EC_{50} values, which were $25.3 \pm 1.44 \mu$ M, $0.6 \pm 0.17 \mu$ M, $2.7 \pm 0.59 \mu$ M, $5.9 \pm 0.63 \mu$ M, and $0.3 \pm 0.12 \mu$ M, respectively. Dose-response analysis revealed that compound **19** displayed the most potent neuroprotective effect, with the lowest EC_{50} ($0.3 \pm 0.12 \mu$ M), indicating markedly higher efficacy compared with the other isolated compounds. Notably, previous studies have not reported the neuroprotective properties of compound **19**, either in general neuronal systems or specifically in retinal ganglion cells or *in vivo* retinal injury models. These highlights compound **19** as a promising candidate for further investigation, supporting its potential for deeper evaluation in both *in vitro* and *in vivo* models.

2.2.2. Antioxidant effects of MAEA and its isolated compounds against glutamate-induced cytotoxicity in R28 cells

Fluorescence analysis showed that glutamate markedly increased intracellular ROS to 2.23-fold of the control ($p < 0.0001$), consistent with the strong DCF fluorescence observed in the glutamate group (Figure S70A). MAEA and the bioactive compounds reduced this elevation in a dose-dependent manner, with fluorescence intensity decreasing as concentrations increased and approaching control levels. At their highest concentrations, MAEA (10 μ g/mL) lowered ROS to 1.22-fold, followed by compound **2** and compound **19** (1.07-fold), compound **9** (1.16-fold), and compound **18** (1.22-fold),

whereas compound **11** remained relatively higher (1.40-fold) (Figure S70B,C). NAC (1 mM) normalised ROS to 0.99-fold. Overall, these results confirm that MAEA and its constituents effectively suppress glutamate-induced ROS generation, demonstrating their antioxidant activities.

2.2.3. Protective effects of MAEA and its isolated compounds against glutamate-induced apoptosis in R28 cells

Hoechst 33342/propidium iodide (PI) staining was used to quantify apoptotic cell populations, with all values normalised to the control (set to 1). As shown in Figure S71A, glutamate (10 mM) markedly increased PI staining, elevating the PI/Hoechst value to approximately 22-fold relative to the control, indicating severe excitotoxic cell death. This increase was substantially reversed by MAEA, which reduced PI-positive nuclei towards baseline levels. Co-treatment with MAEA and the isolated compounds also significantly attenuated glutamate-induced PI elevation in a dose-dependent manner (Figure S71B,C). Among the bioactive constituents, compounds **2**, **9**, **11**, **18**, and **19** exhibited the strongest protective effects, decreasing PI/Hoechst fold-changes towards control levels at their highest concentrations (50, 1, 5, 10, and 1 μ M, respectively) (Figure S71C). These findings demonstrate that MAEA and its constituents effectively limit membrane disruption and late-stage apoptosis, confirming their strong neuroprotective activity against glutamate-induced cytotoxicity.

2.2.4. Neuroprotective effects of compound **19** on retinal structure and function in NMDA-induced injury model

Although several compounds exhibited comparable activities in the initial screening, compound **19** displayed the most potent protective effect, as indicated by its lowest EC₅₀ value among the tested compounds. In addition, compound **19** consistently demonstrated strong neuroprotective effects across multiple *in vitro* assays. Based on these results, compound **19** was selected for further evaluation in an *in vivo* model. Notably, the neuroprotective potential of compound **19**, particularly in the NMDA-induced retinal injury model, has not been previously reported.

Optical coherence tomography (OCT) was performed to evaluate retinal structure and thickness (Figure S72A). OCT revealed that intravitreal *N*-methyl-D-aspartate (NMDA) injection markedly reduced total retinal thickness, particularly in the inner retinal layers, compared with controls (Figure S72A–C). Co-treatment with compound **19** (0.05 mM) significantly preserved retinal thickness ($p < 0.05$). Haematoxylin and eosin (H&E) staining confirmed these findings (Figure S72D), showing that NMDA induced disorganisation of retinal structure, thinning of the ganglion cell layer (GCL) and inner nuclear layer (INL), and reduced cell density, whereas compound **19** reversed these changes. Quantitative analysis showed that NMDA significantly decreased GCL + inner plexiform layer (IPL) and INL thickness and the number of nuclei per 100 μ m of GCL ($p < 0.001$), while compound **19** co-treatment restored these parameters ($p < 0.05$) (Figure S72E–G). Western blot analysis demonstrated that NMDA markedly upregulated Bcl-2-associated X protein and cleaved caspase-3 ($p < 0.0001$), both of which were significantly suppressed by compound **19** ($p < 0.001$) (Figure S73). Consistent with the *in vitro* findings, compound **19** markedly attenuated NMDA-induced retinal

damage, demonstrating its protective effects against excitotoxic retinal injury. However, the current data does not support a role for compound **19** as a direct NMDA receptor antagonist. Instead, its neuroprotective activity appears to be mediated through modulation of downstream pathological processes, including apoptosis, that are triggered by NMDA receptor overactivation. Taken together, these results indicate that compound **19** effectively prevents NMDA-induced retinal degeneration by preserving retinal structure and suppressing apoptosis.

3. Experimental

See [Supplementary Material](#).

4. Conclusions

This study expanded the phytochemical profile of *M. alba* root bark through the isolation of 23 compounds, including two new compounds (**1** and **2**) and six new natural products (**3–8**) occurring as stereoisomeric mixtures. Bioactivity screening identified compounds **2**, **9**, **11**, **18**, and **19** as neuroprotective candidates, with compound **19** (albasin B) exhibiting the strongest activity. Compound **19** effectively mitigated NMDA-induced retinal degeneration by preserving retinal structure and reducing apoptosis. These findings suggest that compound **19** represents a promising lead for further investigation in the development of therapeutics targeting excitotoxic and glaucomatous neurodegeneration. However, several limitations should be acknowledged, including the use of a single acute *in vivo* injury model and the absence of detailed pharmacokinetic and long-term safety evaluations. Future studies should evaluate the neuroprotective efficacy of compound **19** in additional experimental models, together with comprehensive pharmacokinetic profiling and long-term safety assessments to further validate its therapeutic potential.

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Author contributions

CRedit: **Tam Thi Le**: Data curation, Formal analysis, Methodology, Writing – original draft, Writing – review & editing; **Hai Yen Nguyen**: Data curation, Formal analysis, Methodology, Writing – original draft; **Hyejin Ko**: Formal analysis, Methodology; **Hong Ryul Ahn**: Data curation, Methodology; **Trang Thi Thu Pham**: Formal analysis, Methodology; **Sungjae Yang**: Conceptualization, Data curation, Formal analysis, Methodology, Visualization, Writing – review & editing; **Sang Hoon Jung**: Conceptualization, Methodology, Supervision, Writing – review & editing.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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